# How to Choose the Right Stock to Purchase

Lesson 6: Technical Analysis Intro

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## Technical Analysis Intro – Study Notes

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### Key Concepts

1. **Moving Averages (MAs)** – Smoothing tools that help reveal the direction of a price trend.
2. **Support & Resistance (S/R)** – Price levels where buying or selling pressure historically pauses or reverses.
3. **Volume** – The number of shares/contracts traded; confirms the strength of a move or a breakout.
4. **Trend Identification** – Determining whether the market is in an up  trend, down  trend, or ranging (sideways) phase.
5. **Confluence** – Using two or more technical tools together to increase confidence in a signal.

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### Summary

Technical analysis seeks to forecast future price movements by studying past market data—primarily price and volume. For beginners, the most accessible way to spot a trend is by applying moving averages to a chart. A short  term MA (e.g., 20  day) reacts quickly to price changes, while a longer  term MA (e.g., 50   or 200  day) smooths out noise. When the short  term MA crosses above the long  term MA, it often signals the start of an up  trend (a “golden cross”). The opposite crossover, a “death cross,” can indicate a down  trend.

Support and resistance levels act as psychological barriers where traders historically place buy (support) or sell (resistance) orders. Prices tend to bounce off these zones, creating predictable patterns. Volume adds a third dimension: a price move accompanied by high volume is generally more reliable than one on low volume. By watching how price interacts with moving averages, S/R zones, and volume, traders can confirm trends, anticipate reversals, and make more informed entry and exit decisions.

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### Important Points

• **Moving Average Types**: Simple (SMA) = arithmetic mean; Exponential (EMA) = gives more weight to recent prices.

• **Crossovers**:

- **Golden Cross** – Short  term MA crosses **above** long  term MA !’ bullish signal.
- **Death Cross** – Short  term MA crosses **below** long  term MA !’ bearish signal.

• **Support**: A price level where demand overtakes supply; often formed by prior lows or a rising MA.

• **Resistance**: A price level where supply overtakes demand; often formed by prior highs or a falling MA.

• **Volume Confirmation**:

- Rising price + rising volume !’ strong up  trend.

- Falling price + rising volume !' strong down  trend.
- Price move on low volume !' possible false breakout.

• **Trend Direction**:

- **Up  trend** – Higher highs & higher lows; price stays above a rising MA.
- **Down  trend** – Lower highs & lower lows; price stays below a falling MA.
- **Sideways** – Price oscillates between clear S/R levels; MAs flatten.

• **Confluence**: The more tools that agree (e.g., price above MA, bouncing off support, and high volume), the higher the probability of a reliable signal.

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# Examples

### Example 1 – Identifying an Up  Trend with a 20  Day EMA & Volume

1. **Chart Setup**: Apply a 20  day EMA (blue) and a 50  day EMA (red) to a daily price chart of XYZ Corp.
2. **Observation**: On Day 45, the 20  day EMA crosses above the 50  day EMA.
3. **Volume Check**: The crossover day shows a 30 /% increase in daily volume versus the 20  day average.
4. **Interpretation**: The bullish “golden cross” plus strong volume suggests the start of an up  trend. A trader might consider entering a long position with a stop just below the recent swing low (support).

### Example 2 – Using Support, Resistance, and Volume to Confirm a Breakout

1. **Chart Setup**: Identify a well  defined resistance zone at $55 on a 1  hour chart of ABC Inc.
2. **Price Action**: The price approaches $55, tests it twice, and on the third attempt pushes above $55.
3. **Volume Confirmation**: The breakout candle is accompanied by volume 2.5× the average 20  period volume.
4. **Result**: The confluence of a clear resistance break and high volume validates the breakout, indicating a potential continuation higher. A trader could place a buy order just above $55 with a stop below the breakout candle’s low.

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# Quick Review

1. **What is the main difference between a Simple Moving Average (SMA) and an Exponential Moving Average (EMA)?**
2. **Define “golden cross” and “death cross.” What do they typically indicate?**
3. **Why is volume important when confirming a price breakout?**
4. **How can you distinguish an up  trend from a down  trend using higher highs/lows?**
5. **What does “confluence” mean in technical analysis, and why is it valuable?**

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# Further Reading

• **Advanced Moving Averages** – Weighted MA, Hull MA, and adaptive moving averages.

• **Candlestick Patterns** – How price action shapes (e.g., engulfing, pin bar) interact with MAs and S/R.

• **Trend  Following Systems** – The Turtle Trading rules, Donchian Channels, and ATR  based stops.

• **Volume  Based Indicators** – On  Balance Volume (OBV), Volume Profile, and the Money Flow Index (MFI).

• **Risk Management** – Position sizing, stop  loss placement, and the role of risk  reward ratios in trend trading.